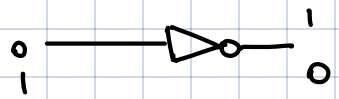
# II. Quantum Circuits

- "circuit model": sequence of building blocks that carry out elementary computations, called gates

## Single qubit gates

- classical example: NOT




- quantum gates: as quantum theory is unitary, quantum gates are represented by unitary matrices

$$U^\dagger U = U U^\dagger = I$$

recall dirac notation

$$U = \begin{pmatrix} u_{00} & u_{01} \\ u_{10} & u_{11} \end{pmatrix} = u_{00} |0\rangle\langle 0| \overset{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}{=} \oplus$$

$$\oplus\ u_{01} |0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\oplus\ u_{10} |1\rangle\langle 0|$$

$$+\ u_{11} |1\rangle\langle 1|$$

### Examples

① $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = |0\rangle\langle 1| + |1\rangle\langle 0|$

↳ $\sigma_x |0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$

↳ $\sigma_x |1\rangle = (|0\rangle\langle 1| + |1\rangle\langle 0|) \cdot |1\rangle \overset{\oplus}{=}$

$\oplus\ |0\rangle \underbrace{\langle 1|1\rangle}_{1} + |1\rangle \underbrace{\langle 0|1\rangle}_{0} =$

$= |0\rangle.$ **So it is a bit flip.**

kinda NOT-gate

③

$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

e.g $|0\rangle - \boxed{\sigma_x} - |1\rangle$

↳ rotation around x-axis by $\pi$.

i. $\sigma_x \cdot \sigma_y \Rightarrow$ bit & phase flip.

$\Rightarrow \sigma_x, \sigma_y$ & $\sigma_z$ are the so-called Pauli matrices and

$\sigma_i^2 = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

if we apply the same gate twice, the result would be nothing.

② $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = |0\rangle\langle 0| - |1\rangle\langle 1|$

↳ $\sigma_z |+\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \overset{\oplus}{=}$

$\oplus\ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle$

↳ $\sigma_z |-\rangle = (|0\rangle\langle 0| - |1\rangle\langle 1|) \cdot \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) =$

$= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = |+\rangle.$

**So it is a phase flip**

↳ rotation around z-axis by $\pi$.

Pauli matrices with identity they form a basis of 2x2 matrices.

④ Hadamard gate:

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}}\left(|0\rangle\langle 0| + |0\rangle\langle 1| + |1\rangle\langle 0| - |1\rangle\langle 1|\right)$$

↳ $H|0\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle.$

↳ $H|1\rangle = |-\rangle$

This creates superposition

Also $H|+\rangle = |0\rangle$, $H|-\rangle = |1\rangle$

used to change between X & Z bases.

⑤ $S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$ adds $90°$ to the phase $\varphi$

$S|+\rangle = |+i\rangle$

$S|-\rangle = |-i\rangle$

⇒ $S \cdot H$ is applied to change from Z to Y basis.

## Multipartite quantum states

$$A \cdot \vec{x} = \lambda \cdot \vec{x}$$

- we use tensor products to describe multi-partite states:

$$|a\rangle \otimes |b\rangle = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \otimes \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_1 b_1 \\ a_1 b_2 \\ a_2 b_2 \\ a_2 b_2 \end{pmatrix}$$

- example: System A is in state $|1\rangle_A$ and system B is in state $|0\rangle_B$ ⇒ the total (bipartite) state is $|10\rangle_{AB} = |1\rangle_A \otimes |0\rangle_B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$

↳ remark: states of this form are called uncorrelated, but there are also bipartite states that cannot be written as $|\psi\rangle_A \otimes |\varphi\rangle_B$. These states are correlated and sometimes even entangled.

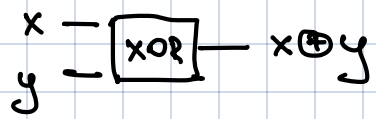
This scales up very quickly!

e.g $|\psi\rangle_{AB} = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle_{AB}\right) =$

$$= \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

# Two-qubit gates

- classical example: XOR




this is an irreversible gate (cannot construct the input from the output)

BUT: as quantum theory is unitary, we only consider unitary and therefore reversible gates.

$$U^\dagger U = I \iff U^{-1} = U^\dagger$$

- quantum example.

$$\text{CNOT} = \begin{array}{c} \begin{array}{cccc} 00 & 01 & 10 & 11 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{array} = |00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 11| \oplus$$
$$\oplus\ |11\rangle\langle 10|$$

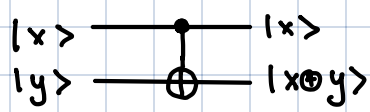
↳ $\text{CNOT} \cdot |00\rangle_{xy} = \text{CNOT} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = |00\rangle_{xy}$

↳ $\text{CNOT} \cdot |10\rangle_{xy} = |11\rangle_{xy}$.

Consider the input output table

| input | output | |
|---|---|---|
| x y | x | x⊕y |
| 0 0 | 0 | 0 |
| 0 1 | 0 | 1 |
| 1 1 | 1 | 1 |
| 1 1 | 1 | 0 |

⇒ circuit




≡ reversible XOR.

⇒ we can show that every function f can be described by a reversible circuit.

⇒ quantum circuits can perform all function that we can calculate classically.